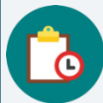


8.1 Proving Properties of Parallelograms

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.		 Learning Targets	- I can define a parallelogram. - I can prove opposite sides and opposite angles of a parallelogram are congruent. - I can prove consecutive angles of a parallelogram are supplementary. - I can prove diagonals of a parallelogram bisect each other.	
 Benchmark Coherence	Building On N/A			Working Toward N/A	
 Essential Questions	- What defines a figure as a parallelogram? - What are the properties of a parallelogram? - How do we prove that a given quadrilateral is a parallelogram?				
 Lesson Narrative	Students explore parallelograms by measuring sides, angles, and diagonals to make conjectures about the properties of parallelograms. Using definitions, postulates, and theorems, students will complete proofs involving parallelograms.				
 Component Summary	8.1.1 Warm-Up (~5 min.)	Review a career video of a Cartographer and respond to questions about maps. <i>(SE p. 7)</i>	8.1.3 Exploration (15-20 min.)	Write conjectures about the properties of parallelograms. <i>(SE p. 8)</i>	
	8.1.2 Collaboration (5 min.)	Draw and classify a four-sided, enclosed shape. <i>(SE p. 8)</i>	8.1.4 Guided Instruction (10-15 min.)	Complete proofs about parallelograms. <i>(SE p. 11)</i>	
	8.1.5 Wrap-Up (~5 min.)	Reflect on parallelograms by providing a fact that was known prior to the lesson, and a new fact that was learned. <i>(SE p. 12)</i>			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Parallelogram <u>Additional Terms:</u> - Kite		- Patty Paper - Ruler - Protractor		8.1.1 Warm-Up contains a video to accompany the question prompts

 Teacher Tips	Common Misconceptions - Students may assume that the measure of all angles of the shape will always be 90° for 8.1.3 Exploration, question 3. - Students may state a segment bisector cuts a segment into congruent parts. The definition of segment bisector states equal parts, not congruent. This is important to note for proper use later in writing proofs.	Things to Consider Consider adding the key information about parallelograms from this lesson to a proof resource booklet, a foldable, or flipbook, if one was started prior to this unit. If not, consider starting one and naming it “parallelograms.”
	Literacy and Language Routines Think-Pair-Share Consider utilizing a Think-Pair-Share routine for 8.1.4 Guided Instruction, questions 2 and 3. This will allow students the opportunity to independently approach each proof, and discuss their thinking before teacher facilitation.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students are positioned in 8.1.3 Exploration, question 5 to develop illustrations highlighting the different properties of parallelograms. This is an opportunity for students to demonstrate understanding of the content through drawing and naming parallelograms, and writing a mathematical statement for the picture they drew.
	 Differentiate Instruction	
Lesson Notes:		

8.1.1 Warm-Up: Cartographer

Component Overview: Play the career profile video on the Cartographer for the entire class. After students watch the video (4:12 minutes), prompt them to reflect on the profession by answering the two questions independently.



8.1.1 Warm-Up: Cartographer

Dave Imus is a cartographer whose maps have won many awards. Dave sees the world through understanding places. Dave's maps are considered art.

1. Have you ever used a paper map?

Answers will vary. No, I use my GPS to find my way to a new place.

2. Since you can't put everything on a map, what three places do you think should be on the map of Florida?

Answers will vary. I think Disney World, Pensacola, and Daytona Speedway.



Consider bringing in different paper maps of Florida or different states, so that students can see and experience a map in person.

8.1.2 Collaboration: Parallel Lines and Four-Sided Figures

Component Overview: Using patty paper, students will create parallelograms from two set of parallel lines. Students will classify the type of parallelogram they created, and justify their classification.



8.1.2 Collaboration: Parallel Lines and Four-Sided Figures

1. On a piece of patty paper, draw one set of parallel lines. Overlap your patty paper with a partner's parallel lines to create an enclosed figure. Each partner should trace the enclosed shape onto a new piece of patty paper.
 - a. Classify the enclosed shape.
parallelogram
 - b. Justify your classification.
A parallelogram is a four-sided shape with opposite sides that are parallel.
 - c. Ask two classmates how they classified the figure they created. Record their classification and write your summary of their justification. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Classification	My Summary of Their Justification	Initials



If needed, refer students back, very briefly, to prior lessons on creating parallel lines. Focus on the definition of parallel lines, and angle relationships formed by parallel lines and a transversal.



Students may draw these lines very close to each other. Remind students to leave at least 10 cm between the lines and continue the lines to the very edge of the paper.

8.1.3 Exploration: Properties of Parallelograms

Component Overview: Using the parallelograms created during the Exploration, students will measure segment lengths and angles to determine the properties of parallelograms. The students will then represent the properties in words, pictorially, and using mathematical statements.



8.1.3 Exploration: Properties of Parallelograms



A quadrilateral containing two pairs of parallel sides is a **parallelogram**. A member of the following shape classes: polygons, quadrilaterals, trapezoids. Sub-classes include rectangles, rhombi, and squares.

1. Using a ruler, measure the pairs of opposite sides in each parallelogram you and your partner created.
 - a. What do you notice?
They are the same length.
 - b. Make a conjecture about the measures of opposite sides of a parallelogram.
Opposite sides of a parallelogram are congruent.
2. Using a ruler, draw the diagonals of one of the parallelograms.
 - a. Determine the lengths of each diagonal.
Answers will vary.
 - b. Mark the point where the two diagonals intersect.
Answers will vary.
 - c. What do you notice about the point of intersection and its relationship to each diagonal?
The point of intersection is the midpoint of both diagonals.
 - d. Repeat the same steps to draw and measure the diagonals in your second parallelogram.
Answers will vary.
 - e. Make a conjecture about the diagonals of a parallelogram.
Each diagonal of a parallelogram bisects the other diagonal.



Students may believe that all the angle measures will always be 90° . Consider asking students to draw the shape differently to see if the angles are still right angles.

8.1.3 Exploration: Properties of Parallelograms (continued)

3. Using a protractor, measure the pairs of opposite interior angles of your parallelograms.
 - a. What do you notice?
The opposite interior angles have the same measure.
 - b. Make a conjecture about the measures of opposite angles of a parallelogram.
The opposite interior angles of a parallelogram are congruent.
 - c. Make a conjecture about the measures of consecutive angles of a parallelogram.
Consecutive angles of a parallelogram are supplementary angles.
4. Based on your conjectures, select all of the statements about a parallelogram that are true.
 - ☐ A parallelogram has only one set of opposite parallel sides.
 - ☐ *A parallelogram has two sets of opposite parallel sides.*
 - ☐ A parallelogram has only one set of opposite congruent sides.
 - ☐ *A parallelogram has two sets of opposite congruent sides.*
 - ☐ Consecutive angles of a parallelogram are congruent.
 - ☐ *Consecutive angles of a parallelogram are supplementary.*
 - ☐ *Opposite angles of a parallelogram are congruent.*
 - ☐ Opposite angles of a parallelogram are supplementary.
 - ☐ *Diagonals of a parallelogram bisect each other.*

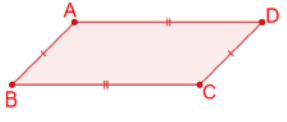
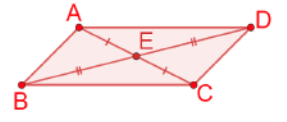


True or False

Consider utilizing this routine when question 4 by displaying one statement at a time and having students indicate their answers using thumbs up/thumbs down or holding up cards. As students share why they selected their answer, prompt them to justify their response, whether right or wrong, with correct mathematical vocabulary. Listen for words and phrases like “transversal,” “distance between lines,” “equal length,” etc.

8.1.3 Exploration: Properties of Parallelograms (continued)

5. For each true statement in the previous problem, draw and name a parallelogram. Using the information in the statement, mark the diagram and write a mathematical statement. The first row has been completed.

Statement	Drawing	Given
A parallelogram has two sets of opposite parallel sides.		$\overline{AD} \parallel \overline{BC}$ $\overline{AB} \parallel \overline{DC}$
A parallelogram has two sets of opposite congruent sides.		$\overline{AD} \cong \overline{BC}$, $\overline{AB} \cong \overline{DC}$
Consecutive angles of a parallelogram are supplementary.		$\angle ABC$ and $\angle BCD$ are supplementary, $\angle BCD$ and $\angle CDA$ are supplementary, $\angle CDA$ and $\angle DAB$ are supplementary, $\angle DAB$ and $\angle ABC$ are supplementary
Opposite angles of a parallelogram are congruent.		$\angle ABC \cong \angle CDA$, $\angle BCD \cong \angle DAB$
Diagonals of a parallelogram bisect each other.		$AE = CE$, $BE = DE$



MA.K12.MTR.2.1: Multiple Representations

The design of this activity allows students to represent the properties of parallelograms in multiple ways. Students begin identifying each statement by the words. Using this, students are invited to develop a pictorial and mathematical representation of each property.



Students may write the mathematical statement using “congruence,” instead of “equality.” Have students recall the definition of segment bisectors from Unit 1 to determine which to use.

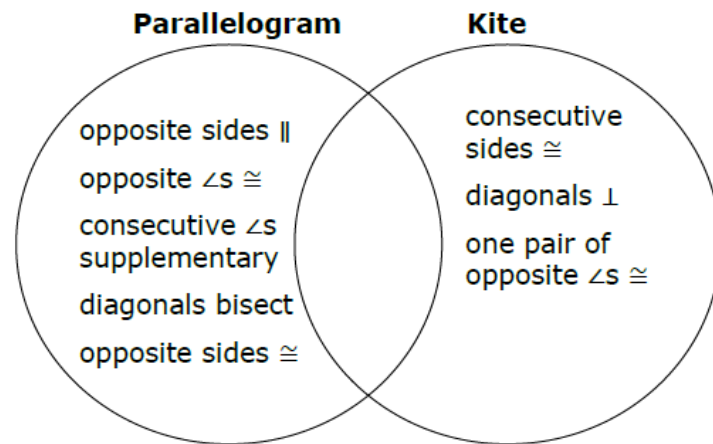
8.1.3 Exploration: Properties of Parallelograms (continued)

6. The Venn diagram shows the properties of a parallelogram and another quadrilateral called a kite. Discuss with your partner why a kite is not considered a parallelogram.



Students have not been exposed to the vocabulary word “kite.” This word is defined in Lesson 4.

Students are not required to write a summary of their discussion. While monitoring student conversation, listen for students to connect that a figure can’t be both a parallelogram and a kite, but could be one or the other.



Properties of Parallelograms

- Opposite sides of a parallelogram are congruent.
- Opposite angles of a parallelogram are congruent.
- Consecutive interior angles of a parallelogram are supplementary.
- The diagonals of a parallelogram bisect each other.

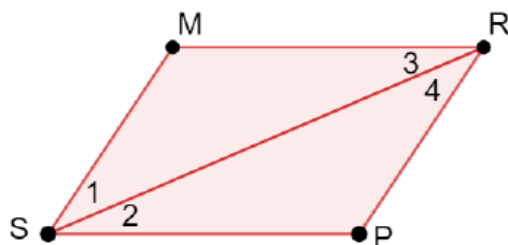
8.1.4 Guided Instruction: Proofs With Parallelograms

Component Overview: Guide students through proving the theorems related to the properties of parallelograms. The intent is to complete these proofs whole-group, allowing students to take the lead in filling in the statements and reasons.



8.1.4 Guided Instruction: Proofs with Parallelograms

1. $MRPS$ is shown.



Given: $MRPS$ is a parallelogram

Prove: $\overline{MR} \cong \overline{PS}$ and $\overline{MS} \cong \overline{PR}$

Complete the following two-column proof.

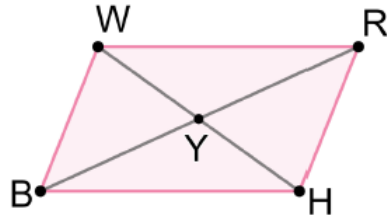
Statement	Reason
1. $MRPS$ is a parallelogram	1. Given
2. $\overline{MR} \parallel \overline{SP}, \overline{MS} \parallel \overline{RP}$	2. Definition of a parallelogram
3. $\angle 1 \cong \angle 4, \angle 2 \cong \angle 3$	3. If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.
4. $\overline{SR} \cong \overline{SR}$	4. Reflexive property of congruence
5. $\triangle MSR \cong \triangle PRS$	5. ASA congruence theorem
6. $\overline{MR} \cong \overline{PS}$ and $\overline{MS} \cong \overline{PR}$	6. CPCTC



As an additional scaffold, consider providing the possible statements and reasons on note cards for students who still struggle with writing proofs to choose from.

8.1.4 Guided Instruction: Proofs With Parallelograms (continued)

2. $WRHB$ with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y is shown.



Given: $WRHB$ is a parallelogram with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y .

Prove: $\overline{WH}$ bisects $\overline{BR}$
 $\overline{BR}$ bisects $\overline{WH}$

Complete the following two-column proof.

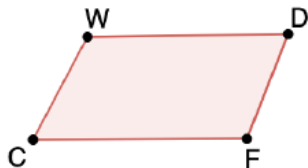
Statement	Reason
1. $WRHB$ is a parallelogram with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y .	1. Given
2. $\overline{WR} \parallel \overline{BH}$	2. Definition of a parallelogram
3. $\overline{RH} \cong \overline{BW}$; $\overline{WR} \cong \overline{HB}$	3. Opposite sides of a parallelogram are congruent.
4. $\angle WRB \cong \angle HBR$; $\angle BHW \cong \angle RWH$	4. If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.
5. $\triangle BYH \cong \triangle RYW$	5. ASA congruence theorem
6. $\overline{BY} \cong \overline{RY}$ and $\overline{WY} \cong \overline{HY}$	6. CPCTC
7. $BY = RY$ and $WY = HY$	7. Definition of congruence
8. $\overline{WH}$ bisects $\overline{BR}$ $\overline{BR}$ bisects $\overline{WH}$	8. Definition of a segment bisector



Students may include both sets of parallel lines in the proof. Facilitate a discussion on why it is not necessary to include $\overline{WB} \parallel \overline{RH}$. Ask probing questions like, "What angles are congruent by stating these two lines are parallel? Are these angles needed to prove $\triangle BYH \cong \triangle RYW$?"

8.1.4 Guided Instruction: Proofs With Parallelograms (continued)

3. Complete the following flowchart proof. Statements go in the boxes and reasons go below the boxes.

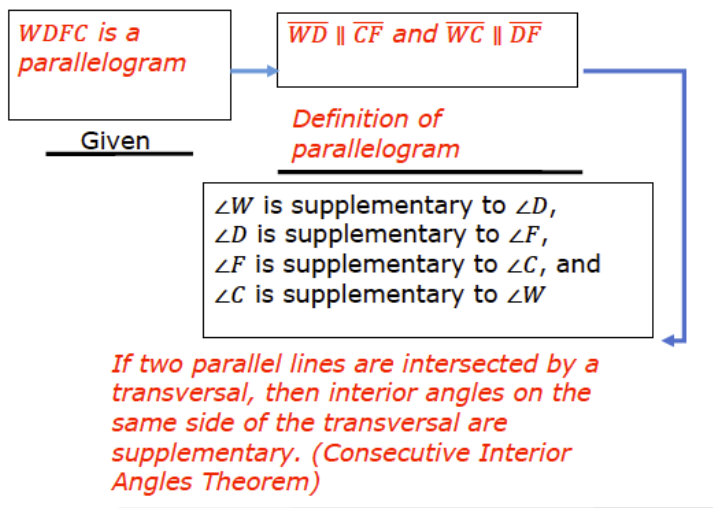


Given: $WDFC$ is a parallelogram.

Prove: $\angle W$ is supplementary to $\angle D$,
 $\angle D$ is supplementary to $\angle F$,
 $\angle F$ is supplementary to $\angle C$, and
 $\angle C$ is supplementary to $\angle W$



Consider allowing them to complete question 3 as a Think-Pair-Share.



8.1.5 Wrap-Up: Cool Down

Component Overview: Students reflect on what they knew already about parallelograms, if anything, and identify something new that they learned.



8.1.5 Wrap-Up: Cool Down

1. Write down one fact you knew about parallelograms before the lesson.

Answers will vary.

2. Write down one fact you learned about parallelograms during the lesson.

Answers will vary.



If there is extra time, ask students to form their answers to question 2 into assessment questions. Ask students to write these questions on note cards, and use them in a future review lesson.